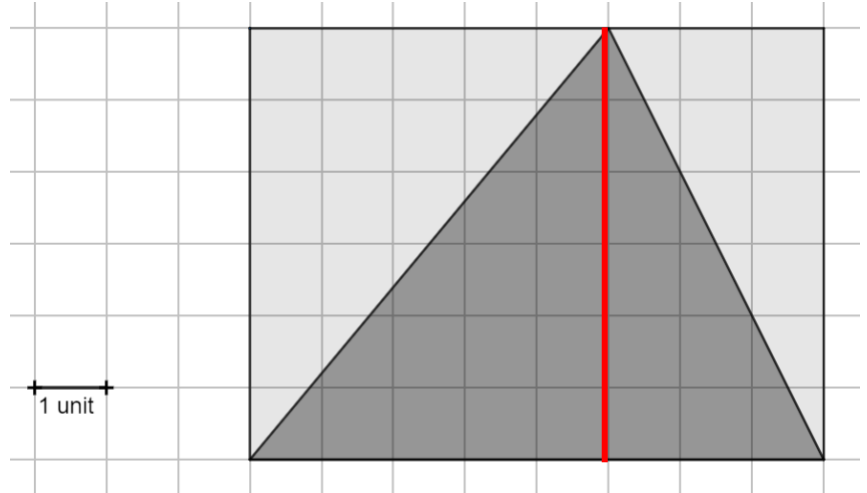


Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result.

1. A rectangle that has a length of 8 units and a width of 6 units with a triangle that is inscribed in the rectangle is shown.



- a. Draw a line segment that shows the height of the triangle.
- b. Determine the area of the triangle.

Area

24 square units

- c. Complete the statements.

The triangle and the rectangle have

- ☐ only the same base.
- ☐ only the same height.
- ☒ both the same base and the same height.

The area of

- ☐ all triangles is
- ☐ any acute triangle is
- ☐ any equilateral triangle is
- ☒ any triangle that is inscribed in a rectangle is

equal to

- ☐ bh ,
- ☐ $b + h$,
- ☒ $\frac{1}{2}bh$,
- ☐ $\frac{1}{2}b + \frac{1}{2}h$,

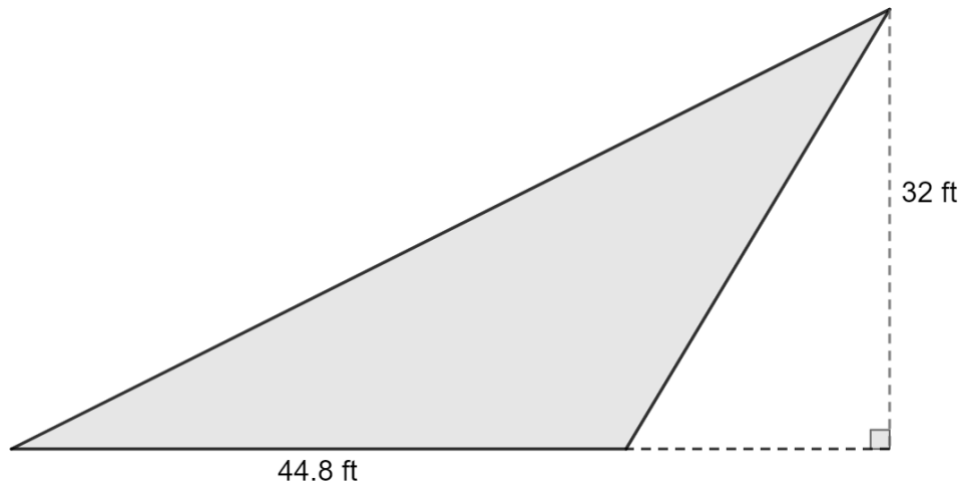
where b

represents the base of the triangle and h represents the height of the triangle.

**Students who select, "all triangles is" for the second answer choice should also be marked correct.*

2. A city planner is designing a skateboard park. The park is triangular in shape. The outline of the park is shown with dimensions in feet (ft).

Determine the area, in square feet, of the skateboard park.



work	
Area	716.8 square feet

3. The volume of a right rectangular prism is 382.5 cubic inches. The height of the prism is 12 inches (in.) and the length of the prism is 7.5 in.

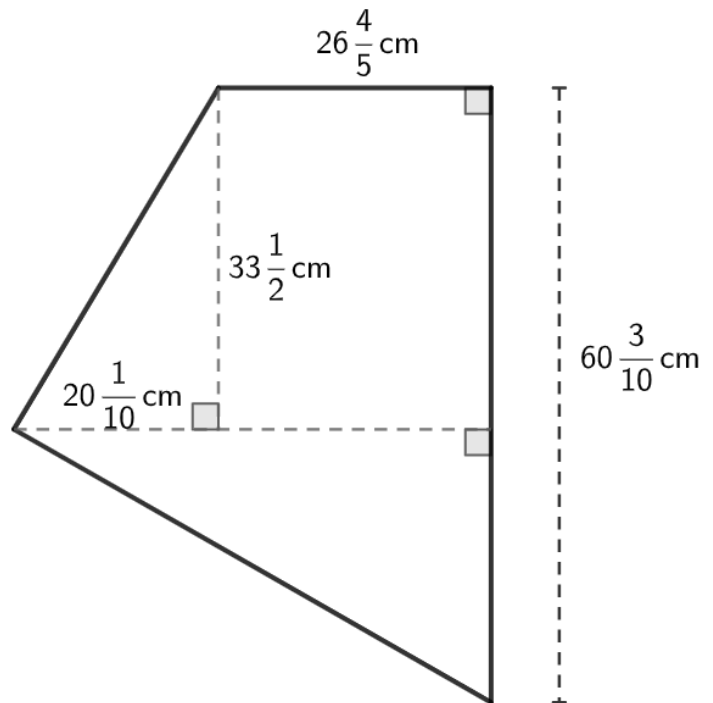
Determine the width, in inches, of the prism.

work	
Width	4.25 inches

4. A quadrilateral with dimensions in centimeters (cm) is shown.

Determine the area, in square centimeters, of the quadrilateral.

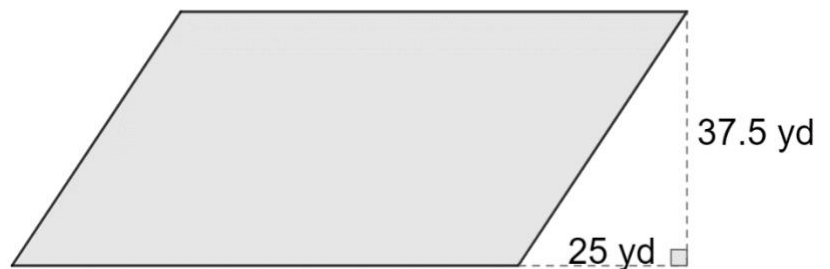
work	
Area	1862 $\frac{187}{200}$ or 1862.935 square cm



5. A quadrilateral with some dimensions, in yards (yd), is shown.

The area of the quadrilateral is 2812.5 square yards.

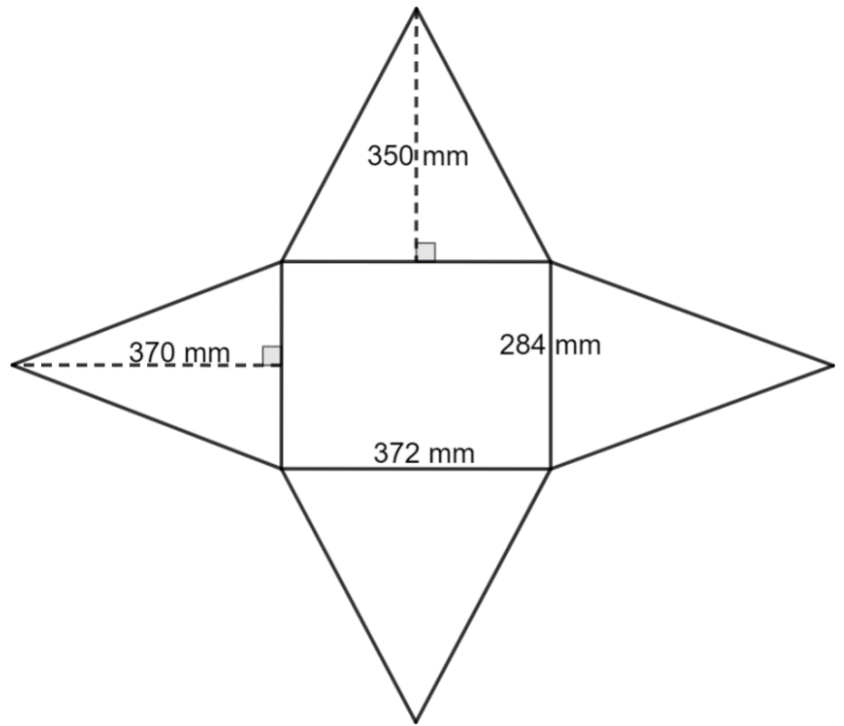
Determine the quadrilateral's missing dimension.



work	
Missing Dimension	75 yards

6. The net of a right rectangular pyramid with dimensions in millimeters (mm) is shown.

Determine the surface area, in square millimeters, of the right rectangular pyramid.



work

Surface Area

340,928 square millimeters

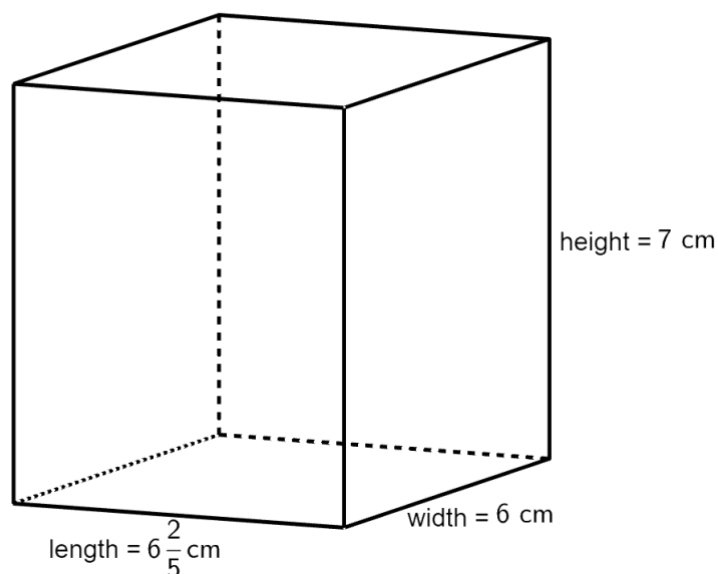
7. A rectangular prism with dimensions in centimeters (cm) is shown.

a. Complete the statements.

To find the volume of the prism,

cubes with side lengths of

- ☐ $\frac{1}{2}$ cm
- ☒ $\frac{1}{5}$ cm
- ☐ $\frac{1}{6}$ cm
- ☐ $\frac{1}{7}$ cm



can be used so that the length will have 32 cubes, the width will have

30 cubes, and the height will have 35 cubes. This results in

33,600 cubes, each with a volume of

- ☐ $\frac{1}{8}$
- ☒ $\frac{1}{125}$
- ☐ $\frac{1}{216}$
- ☐ $\frac{1}{343}$

cubic centimeters.

b. Determine the volume, in cubic centimeters, of the right rectangular prism.

work	
Volume	$268\frac{4}{5}$ or 268.8 cubic centimeters